

Full length article

Evaluation of eco-driving performance of electric vehicles using driving behavior-enabled graph spectrums: A naturalistic driving study in China



Hui Zhang^{a,b}, Yiyue Luo^{a,b}, Naikan Ding^{a,b,*}, Toshiyuki Yamamoto^c, Chenming Fan^{a,b}, Chunhui Yang^{a,b}, Wei Xu^{a,b}, Chaozhong Wu^{a,b,d}

^a Intelligent Transportation Systems Research Center, Wuhan University of Technology, Wuhan 430063, China

^b Engineering Research Center of Transportation Information and Safety, Ministry of Education of the PR China, Wuhan 430063, China

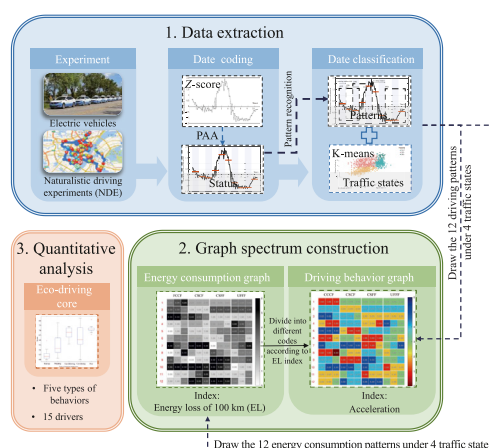
^c Institute of Materials and Systems for Sustainability, Nagoya University, Nagoya 4648603, Japan

^d Hubei University of Arts and Science, Xiangyang 441053, China

HIGHLIGHTS

- Eco-driving behavior performance of electric vehicles were evaluated.
- Driving behavior graph spectrum and energy consumption graph spectrum were constructed via naturalistic driving data.
- Associations between driving behavior and energy consumption under traffic status were decoupled using graph spectrums.
- Energy-intensive drivers accelerate and decelerate significantly more in congested close car-following status.

GRAPHICAL ABSTRACT



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ABSTRACT

Electric vehicles are widely embraced as a promising solution to reduce energy consumption and emission to achieve the Carbon Peak and Carbon Neutrality vision, especially in developing countries. Specifically, it's vital important to understand the ecological performance of electric vehicles and its association with driving behaviors under varying road and environmental conditions. However, current researches on ecological driving behavior mostly use structured data to reflect the characteristics of ecological driving behavior, and it is difficult to accurately reveal the recessive relationship between driving behavior and energy consumption. One promising and prevalent method for comprehensively and in-depth characterizing driving behaviors is "graph spectrums", which allows for an effective and illustrative representation of complex driving behavior characteristics. This study presented an assessment method of ecological driving for electric vehicles based on the graph. Firstly, a multi-source refined data set was constructed through naturalistic driving experiments (NDE). Four typical traffic state (CCCF: congested close car-following; CSSF: constrained slow free-flow; CSCF: constrained slow car-following; UFFF: unconstrained fast free-flow) were classified through longitudinal acceleration data, and

* Corresponding author.

E-mail address: nkding@whut.edu.cn (N. Ding).

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driving behavior graph was constructed to realize the visual representation of driving behavior. Then, the energy consumption graph was constructed using the energy loss of 100 km (EL) index. After the six drivers with the highest and lowest ecological assessment of driving behavior using the behavior graph and energy consumption graph, proposing the quantitative analysis of fifteen drivers' ecology driving behavior. The results show that: 1) The graphical method can describe the individual features of a driver's ecological driving behavior; 2) Rapid acceleration of driving behavior leads to high energy consumption; 3) In the comparison among the six eco-drivers and energy-intensive drivers, founding that the energy-intensive drivers accelerate and decelerate significantly more in CCCF traffic state; 4) The driving behavior was more complex and unecological in CCCF traffic state; 5) Fifteen drivers had lower ecological scores in start-up driving. This study proposes a method for visualizing ecology driving behavior that not only help understand the individual characteristics of ecological driving behaviors, but also offers substantial application value for the subsequent construction of Ecological driving behavior regulation models.

1. Introduction

Emissions of pollutants from transportation accounted for 22 percent of overall pollution in China [1], and vehicular exhaust emissions had become a significant contributor to air pollution. Under such pressure, eco-driving gradually became a vital measure for energy conservation and emission reduction [2,3]. Eco-driving optimization guidance was even included into transportation and traffic related regulations in many countries. Research found that energy-efficient driving behavior could reduce vehicle fuel consumption by at least 5–10% [4], which encourage many countries to embrace the concept of eco-driving, and to develop eco-driving related policies and regulations in recent years [5]. Previous eco-driving studies mostly focused on conventional fuel vehicles. A shift of eco-driving research to electric vehicles are of great importance. According to the recent government statistics, there were over 18 million New Energy Vehicles (NEVs) on roads in China, which accounted for 5.5% of the total number of vehicles. Due to the enormous potential of electric vehicles to reduce emissions and energy consumption, conducting eco-driving research on electric vehicles was a crucial approach for achieving energy conservation, emission reduction, and structural optimization in transportation [6,7].

The performance of eco-driving is influenced by multiple factors, wherein driver behavior is believed to be the most direct and influential one [8]. By better recognizing the characteristic patterns of eco-driving behavior, revealing the coupling relationship between driving behavior and energy consumption and speed, it is possible to enhance the ecological driving performance of electric vehicles. However, the multidimensionality, complexity, randomness, and uncertainty of driving behavior within road conditions, vehicle characteristics, operational states, and others pose challenges in describing and optimizing eco-driving behavior. Traditional statistical methods were used to identify the basic characteristics of eco-driving behaviors, yet still struggled in completely and deeply exploring the complex interactive relationships involved, making it difficult to construct effective and readily feasible measure to reach a well-optimized eco-driving guidance and management [9]. To this end, more effective methods are needed to accurately identify and demonstrate the characteristic patterns of driving behavior. Due to its significant advantages in information visualization, the graph theory had been receiving extensive attention widely and accepted to be an effective and illustrative mean of characterizing the eco-driving with multidimensional data. The driving behavior model established by using the graph theory can visually display the characteristics of driving behavior and quantify the characteristics of different driving behaviors, which could be beneficial and convenient to reveal the behavioral discrepancies and to lead to a better understanding of variations in eco-driving [10].

This study aims to carry out ecological analysis of the driving behavior of electric vehicle drivers, and the driving behavioral data of electric vehicles was collected through a series of naturalistic driving experiments on real-world roads. Particularly, the driving behavior graphs under varying conditions were depicted based on the collected driving behavioral data, the eco-driving behavior and ecological

performance were identified and characterized by using the illustrative methods of graph spectrum, and the spatial-temporal variations of the ecological characteristics of electric vehicles coupled with the variations in driving behavior and traffic flows were further explored. Eventually, the energy consumption graph spectrum was portrayed and the eco-driving performance of electric vehicles was quantified.

2. Literature review

2.1. Effect of ecological driving behavior

Ecological driving is a collection of measures to improve ecological efficiency of the transportation system by changing the driving behaviors of vehicles and its basic idea is to reduce stop-and-go, start, idling, and harsh acceleration or deceleration behaviors. First eco-driving project was conducted by several European countries in 2007, based on the results of the study, suggesting drivers driving in maintain steady speed and smooth acceleration and deceleration [11]. Matthew et al. [12] first proposed the concept of dynamic ecological driving, in which the driver can obtain the surrounding traffic situation in real time, an algorithm for deriving the recommended set speed for these vehicles was described, based on real-time measurements in traffic condition. The simulation results showed that when the traffic flow around the vehicle was relatively small, the energy consumption can be effectively reduced by 10–20 %. Zhao et al. [13] analyzed the influence of driving behavior on vehicle energy consumption under different traffic conditions on urban road, the results showed that under the acceleration conditions had the best effect in ecological driving, and the comprehensive energy saving potential of ecological driving could reach 11.18%. And the influence of ecological driving behavior on energy consumption under different conditions should be considered comprehensively in implementing ecological driving strategies. Wu et al. [14] aimed at the strategy of ecological driving training, conducting a one-month survey of drivers under before and after taking ecological driving training, the results showed that the ecological driving behavior can reduce energy consumption at least by 9–31%. And found that the energy-saving effect of ecological driving for drivers with longer driving experience was worse than the drivers with shorter driving experience, while the driving habits of such longer driving experience drivers were difficult to change.

2.2. Assessment of ecological driving behavior

The key to evaluating the ecological of driving behavior lies in uncovering the implicit correlations between driving behavior and energy consumption emissions. For the analysis of ecological driving behavior characteristics, driving behavior parameters and vehicle engine parameters are the most considered factors. Rolim et al. [15] employed acceleration, deceleration, speed, gear information, and coasting distance to characterize ecological driving behavior. Beusen et al. [16] integrated acceleration, deceleration, speed, gear-shifting characteristics, coasting distance, average engine speed, and throttle/brake pedal travel distance to evaluate the eco-driving performance. Dogan et al. [17] used

indicators such as speed, engine speed, deceleration, braking force, accelerator pedal travel, and brake pedal travel to portray the ecological driving behavior characteristics when vehicles entered and exited intersections. In addition to considering the parameters of the vehicles, the influence of external factors was used, including the signal status and other traffic conditions. Kamal et al. [18] constructed ecological driving behavior models under changing road traffic environmental conditions. Chen et al. [19] derived the optimal driving trajectory of vehicles at intersections with a control objective of the minimum vehicle energy consumption emissions, based on the signal states, the queue length, and driving behavior characteristics.

It can be discovered that numerous indicators can characterize the implicit correlations between driving behavior and energy consumption emissions. Commonly used indicators include vehicle-level parameters such as speed, acceleration, deceleration, gear information, pedal information, etc. Some scholars further considered factors such as road conditions, intersection signal light states, etc. Cheng et al. [20] found that the long idling time and abrupt deceleration behaviors had a great impact on energy consumption emissions. Avoiding these two behaviors can lead to 2.6% and 3.8% decreases on vehicle's energy consumption, respectively. Wang et al. [21] discovered that the fuel consumption was relatively low when the vehicle speed was controlled at 25 km/h and the acceleration was within 0.1–0.4 m/s². Baker et al. [22] and Ando et al. [23] found that rapid acceleration, abrupt acceleration, rapid deceleration, abrupt deceleration, and prolonged idling affect energy consumption emissions greatly.

2.3. Modeling of ecological driving behavior

Several studies have attempted to model driving behavior to characterize its features, achieving acceptable model accuracy [24–26]. However, these studies primarily focused on driving behavior identification, classification, and prediction through mathematical models and statistical analysis [27,28]. The methods for expressing eco-driving behavior characteristics were not effective and illustrative enough to show differences among various drivers. Visual description methods were needed to intuitively describe changes in driver characteristic.

In recent years, the graphical model has been introduced to except to directly reflect the driving characteristics. Using graph theory to establish driving behavior models allows the intuitive presentation of driving behavior features, highlighting differences among various driving behaviors. Through the processing and calculation of graph parameters, driving behavior features can be quantified. Multiple studies have highlighted the applicability of graph theory in characterizing eco-driving behavior [29–31]. Chen et al. [32] proposed a method to establish a driving habit graph (DHG), describing driving habits through the composition of nodes and connecting lines. DHG can illustrate significant changes in behavioral parameters within a series of driving data. However, the DHG may not be appropriate for huge datasets because it requires subjective judgment when drawing. Based on the assumption that drivers have specific driving habits, Chen et al. [33] extracted individual driving behavior patterns and proposed a driver behavior risk features graph using an aggregate approximation algorithm as the expression method. Chandra et al. [34] integrated graph theory with machine learning to predict driving behavior, demonstrating that graph theory-based predictions outperformed traditional data-driven approaches while offering enhanced visualization capabilities.

Introducing graph theory into eco-driving studies can achieve better modeling and feature representation. However, the methods mentioned above for graphical modeling focused on using nodes and lines to present the variation of the critical behavioral parameters in consecutive a few seconds. This could still be limited in that the nodes and lines may not easily cover all the behavioral variations and could largely neglect microscopic yet critical information characterizing eco-driving behaviors. On possible solution could resort to the heap maps, as heat maps are able to accommodate relatively large datasets, discover relationships

between data, and facilitate comparisons between datasets [35]. This approach allows for an intuitive and clear representation of the relationship between driving behavior patterns and energy consumption, enabling the assessment of eco-driving behavior.

3. Experiment and data collection

3.1. Participants

To fully consider the randomness and diversity of driving processes, naturalistic driving experiment was carried out. In the naturalistic driving experiment in this study, 20 normal drivers were selected for the experiment following the below criteria:

- Holding a valid driver's license and with over five years driving experience.
- Be in good health and without any history of major diseases in the recent 5 years.
- Without bad habits such as alcoholism that could impede driving performance.

According to the above screening conditions, the experiment included 16 male drivers and 4 female drivers. The participants ranged in age from 31 years to 58 years, and the driving experience in age from 12 to 33 years.

3.2. Equipment

In this study, an electric vehicle with a nickel-cobalt-manganese lithium battery from Dongfeng was selected (see Fig. 1).

This study improves the data frequency of the collected vehicle data by reanalyzing the CAN data from the on-board terminal device T-BOX and installing corresponding driving behavior data acquisition devices. The data sampling frequency for driving behavior was no less than 10 Hz, while the data sampling frequency for the power battery was no less than 1 Hz. The detailed description of the experimental equipment is presented in Fig. 2 and as follows:

- 1) T-BOX on-board intelligent terminal. Through the electric vehicle OBD interface, the on-board intelligent terminal T-BOX obtains the CAN bus data of the vehicle. After parsing the data using the DBC protocol documentation, real-time monitoring data of the electric vehicle are generated. This intelligent terminal is based on the national standard GB/T 32960 for battery vehicle data fields.
- 2) ADAS cameras equipment. This camera can collect distance headway, time headway, lane line distance and continuous video data in actual traffic environment.
- 3) Three-axis gyroscope. The three-axis gyroscope is primarily used to collect vehicle acceleration, pitch angle, pitch angle acceleration, roll angle, roll angle acceleration, yaw angular speed, and other motion state and vehicle attitude data. The main function of this measurement module is to output three-axis acceleration data, three-axis angular speed data, and three-axis angles.
- 4) Mobile hard disk. After obtaining high-frequency data using the data acquisition devices, it is necessary to store the collected data. Data storage is a crucial step for subsequent data analysis and research. To ensure the stability and reliability of data storage, in this experiment, a Western Digital notebook hard disk with a capacity of 1 TB is used to store the vehicle operation data.

3.3. Data collection and pre-process

This experiment was chosen to conduct from January to September in 2022 in Wuhan city. The experiment involved gathering high-precision battery parameters (including current, voltage, state of charge, and temperature) and driving behavior parameters (such as speed,

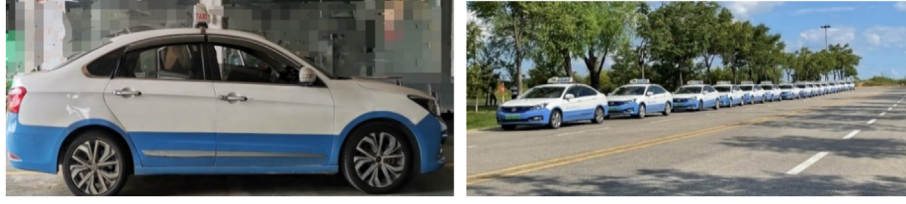


Fig. 1. Experimental vehicles.

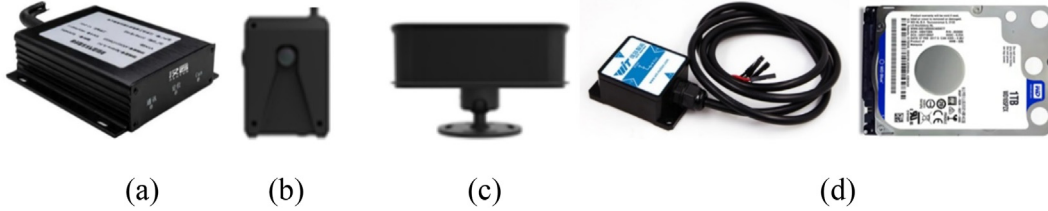


Fig. 2. Testing equipment. (a) Telematics BOX. (b) ADAS cameras equipment. (c) Three-axis gyroscope. (d) Mobile hard disk.

acceleration, brake pedal position, and accelerator pedal position) using various equipment. Additionally, in accordance with the guidelines specified in the national standard GB/T 32960.3-2016 “Technical Specifications for Remote Service and Management System of Electric Vehicles - Part 3: Communication Protocol and Data Format,” this experiment collected data on driver behavior and vehicle status. Table 1 provides details on the specific fields of information collected.

After data processing, the refined data at each timestamp are shown in Table 2.

4. Methodology

Evaluation of eco-driving performance of electric vehicles using graph spectrums Including the following steps: (1) Data collection; (2) Data normalization; (3) Behavior coding; (4) Driving pattern recognition; (5) Traffic state classification; (6) Driving behavior graph construction; (7) Energy consumption graph construction; (8) Quantitative score of ecology driving behavior. Fig. 3 illustrates the above five steps and demonstrates critical processes and results especially showing the graph spectrum construction.

4.1. Data extraction

4.1.1. Data normalization

For the vehicle longitudinal acceleration time series data, the data need to be reasonably simplified to reduce the dimensional differences between different data and to better compare the driving behavior data of various drivers. And this normalization is required in PAA in preparation for the data classification and coding. The acceleration raw data are standardized, and the Z-score method was used to process the ac-

celeration raw data, so that the data of all drivers have the same dimensions, and the Z-score formula is shown in Eq. (1).

$$x^* = \frac{x - \mu}{\sigma} \quad (1)$$

where x is the raw data of vehicle acceleration; μ and σ are the mean and standard deviation of x , respectively; and x^* is the normalised data with a mean of 0 and a standard deviation of 1. The normalised acceleration time-series data is shown in Fig. 4.

4.1.2. Driving pattern recognition

For high frequency time series data, the extracted feature data need to be reasonably simplified to reduce the complexity of the data. Firstly, the data was simplified using Piecewise Aggregate Approximation (PAA) algorithm. Piecewise Aggregate Approximation is a very classical algorithm for dimensionality reduction of time series data. Assuming that the original time series data is $C = \{x_1, x_2, \dots, x_N\}$, the sequence of PAA is defined as $\bar{C} = \{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_w\}$, and the PAA algorithm is processed as shown in Eq. (2).

$$\bar{x}_i = \frac{w}{N} \cdot \sum_{j=\frac{N}{w}(i-1)+1}^{\frac{N}{w}i} x_j \quad (2)$$

where x denotes the original time series data; $\bar{x}$ denotes the PAA-processed sequence data; N denotes the total value of the original time series data; w denotes the total value of the PAA-processed time series data; j denotes the original time data serial number; i denotes the PAA sequence data serial number.

The acceleration data was subjected to PAA to make each frame be 1 s and divided into 7 classes with reference to the division criteria employed previously [33], as shown in Fig. 5. The 7-class division of the normalized acceleration was tested to be the best option against the other ones of 3-, 5-, 9- and 11-classes. Code ‘3’ represented the neutral behaviors. Taking the acceleration data as an example, code ‘3’ represented acceleration values close to 0 m/s². The code “3” to “6” indicating positive acceleration with increased speed, while the code “3” to code “0” suggested the negative acceleration (deceleration) with decreased speed. Since the sampling frequency of the preprocessed data was 10 Hz, so every 10 data points were combined into one PAA data frame.

For a more intuitive representation of driving behavior, a driver’s driving behavior can be decomposed into driving patterns temporally. The key to constructing a driving behavior graph lies in extracting these typical driving patterns. The driving behavior pattern was defined as the driving behaviors during every 3 s, referring to the definition standard of

Table 1

Data items were collected in the experiment.

Data category	Data content	Range of valid values
Vehicle data	Total voltage/V	0~1,000
	Total current/A	-1,000~+1,000
	SOC/%	0~100
	Mileage accrued/km	0~999,999.9
Driving behavior data	Speed/(km·h ⁻¹)	0~220
	Acceleration/(m·s ⁻²)	Unlimited
	Accelerator pedal travel value/%	0~100
	Brake pedal travel value/%	0~100
... ..		

Note: This table omits some field parameters that are not studied in this paper.

Table 2
Time series data.

Timestamp	Acceleration/(m·s ⁻²)	Acc.pedal travel/%	SOC/%	Battery pack voltage/V	Total battery current/A	Speed/(km·h ⁻¹)	...
12:31:47.60	-0.196	36	89.8	384.0	39.0	57.640	...
12:31:47.70	0.098	35	89.8	384.0	37.2	57.768	...
12:31:47.80	0.098	35	89.8	384.2	35.4	57.992	...
12:31:47.90	0.392	35	89.8	384.2	34.6	58.050	...
12:31:48.00	-0.196	34	89.8	384.0	34.4	58.086	...
...	...	...	...	...	...	...	...
12:31:48.80	0.392	33	89.8	384.4	33.0	58.216	...
12:31:48.90	0.098	32	89.8	384.4	33.0	58.231	...

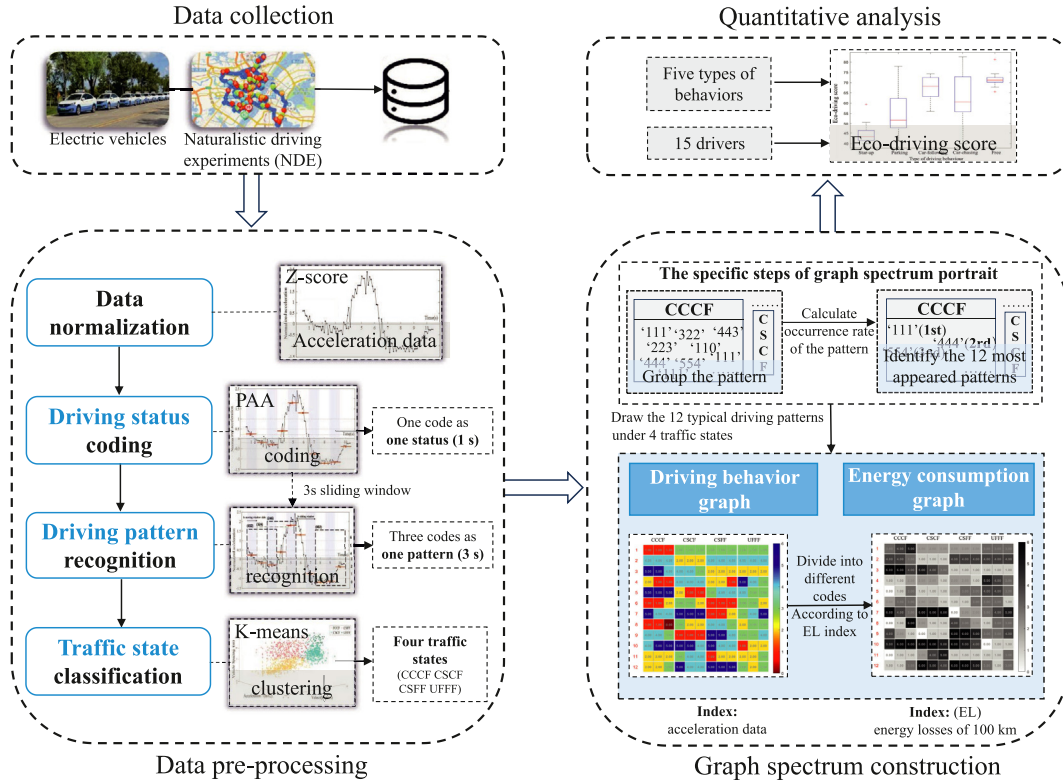


Fig. 3. Steps of the proposed method.

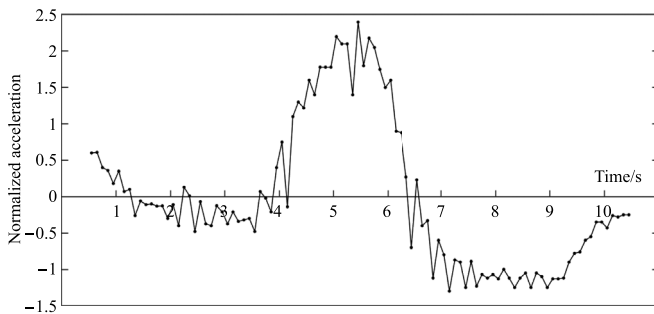


Fig. 4. Acceleration data after standardization.

Chen et al. [33]. In this paper, one individual code is the normalized acceleration value for a time, which was considered as one driving status (1 s). Three consecutive codes comprised one driving pattern (3 s). Be in line with the previous method of digitizing and coding critical driving behaviors [33], in this paper, acceleration or deceleration characteristics in three consecutive seconds were utilized to construct one driving pattern. To better accurately capture the variations of driving patterns across varying drivers, the status was detected under a 3 s moving window size and 1 s moving window slide, as shown in Fig. 5. The detected

driving pattern were “432”, “324”, “246”, ..., “002”. For an example, the pattern “002” suggested a short-term acceleration process as the codes rise from “0” to “2”, suggesting increased speed. Therefore, with these rules, a dataset in n seconds with continuous speed and acceleration can produce $n-2$ patterns.

4.2. Typical traffic states classification

Since drivers' driving behaviors vary greatly in different traffic states, this study first divided the local traffic states by K-Means clustering based on speed and distance headway, and then analyzed the driving behaviors in each local traffic state separately. For the division of traffic flow categories, through the comparison, the $K = 4$ was chosen for the K-Means clustering in this study to possibly and appropriately in accordance with the real-world traffic states. The average speed and the average distance headway within 3 s were used to characterize and classify the local traffic states. Fig. 6 presents the four traffic flow states clustered by K-Means subject to the normalized speed, distance headway, and acceleration.

From Fig. 6, it can be seen that the data points with smaller speed and distance between cars in front of them were grouped into one category, and after observing the ADAS video, it was found that the local traffic condition of these data points was more congested, with closer car-

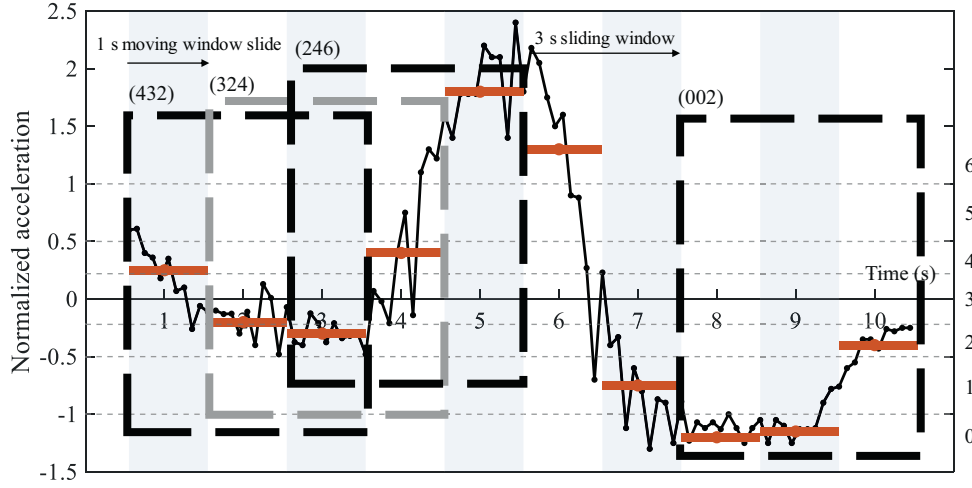


Fig. 5. Driving pattern recognition.

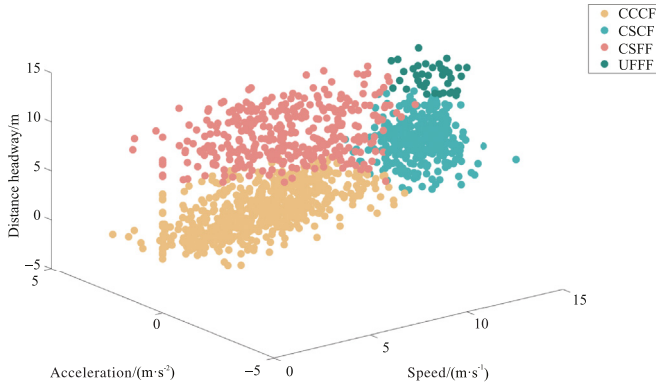


Fig. 6. Traffic states clustering. CCCF: congested close car-following. CSCF: constrained slow car-following. CSFF: constrained slow free-flow. UFFF: unconstrained fast free-flow.

following distance, so the traffic state of this category was named as “congested close car-following (CCCF)”. The data points with moderate speed and small distance between the front vehicles were grouped into one category, and after observing the ADAS video, the local traffic condition of these data points was found to have moderate congestion, moderate car-following distance, smooth and slow vehicle travelling, so the local traffic state of this category was named as “constrained slow car-following (CSCF)”. The data points with small speed and large distance between the front vehicles were grouped into one category, and after observing the ADAS video, it was found that the local traffic condition of these data points was relatively smooth, the car was very little affected by the front vehicle, so the local traffic state of this category was named as “constrained slow free-flow (CSFF)”. The data points with larger speeds were grouped into one category, and after observing the ADAS video, it was found that the local traffic condition of these data points is smoother, the vehicle is in high speed, so the local traffic state of this category was named as “unconstrained fast free-flow (UFFF)”.

4.3. Graph spectrum construction

4.3.1. Driving behavior graph

As introduced previously, the heat map is capable of more effectively accommodating large datasets, capturing relationships between variables within the datasets, and facilitates much more illustrative comparisons between various driving behaviors and behaviors of different driver [35], so the heat map was used for driving behavior graph construction. Based

on the driving pattern extracted in the previous step, the most frequent typical traffic states in various local traffic conditions were used to represent the driving habits of drivers, the longitudinal acceleration is chosen, the specific steps of driving behavior graph spectrum portrait were as follows:

- (1) Cluster driving pattern under the same localized traffic state into one group respectively, and then count the number of individual patterns contained in each group.
- (2) Calculate the occurrence rate of the driving pattern. For an example, for a continuous acceleration dataset of length n in a traffic state, if a driving pattern occurs m times, the occurrence rate of the driving pattern in the traffic state is $m/(n - 2)$.
- (3) Identify the 12 most appeared driving patterns of each driver in each local traffic state, and label them as their typical patterns.
- (4) Draw the typical driving patterns using different colors to represent different driving pattern codes, and sequence the driving pattern in a descending order according to their occurrence rate as in the above steps.
- (5) Combine the heat maps under varying local traffic states, and resulting the complete heat map of a driver showing its acceleration or deceleration driving habits.

By employing the above steps, the driving behavior data can be translated into a two-dimensional graph indicating driving behavioral variations under varying traffic states.

4.3.2. Energy consumption graph

(1) Energy consumption Index

To draw the energy consumption graph, the calculation index of energy consumption should be determined. Considering electric vehicles generate energy feedback when decelerating, only the energy changes of the power battery cannot be evaluated by the pattern's energy consumption, so this paper took into account the energy changes of the power battery and the vehicle's kinetic energy changes of the energy loss (EL) of 100 km to comprehensively evaluate vehicles' energy consumption, as given in Eq. (3) as follows:

$$EL = \frac{100 \times abs \left[\int_{t_1}^{t_2} (UI - P_{AC}) dt - \frac{3.6 \cdot (mv_2^2 - mv_1^2)}{10^6} \right]}{\int_{t_1}^{t_2} v dt} \quad (3)$$

where EL is the energy loss per 100 km for each driving status; U is the output voltage of the power battery; I is the current at the output of the power battery; P_{AC} is the power of the air conditioner; m is the sum of the Experimental vehicle's overall mass of 1,550 kg and the driver's mass of 70 kg; t_1 is the start time of 1s duration; t_2 is the termination time of 1s duration; v_2 is the end speed at 1s time duration; v_1 is the initial vehicle speed at 1s time duration; v is the instantaneous speed of the vehicle over a period of 1s.

(2) Coding rules for energy consumption graph

The 100 km energy loss (EL) of the driving status in all local traffic state was Z-score normalized so that the data for all drivers have the same dimensions. The average 100 km energy loss per second of the standardized driving patterns was divided into seven categories according to the distribution of the size of the 100 km energy loss (see Table 3), to correspond to the coding method for driving behavioral graph spectrum and to possibly associate driving behaviors and energy consumption of vehicles.

Based on the proposed energy consumption evaluation index and the coding rules for energy consumption, the energy consumption graph could be depicted using the same method for the driving behavior graph construction.

4.4. Quantification of ecology driving behavior

According to the previous analysis of the coupling relationship between driving behavior graph, energy consumption graph. Considering that vehicle start-stop behaviors significantly impact energy consumption, particularly pronounced in electric vehicles with more evident acceleration and deceleration feedbacks, the start-up and stopping were used as two behaviors for ecological scoring. Moreover, focusing on free-driving and car-following, two typical driving conditions, and in conjunction with four traffic states, three characteristic driving behaviors were used for ecological scoring. So, this study specifically separated driving behaviors into five types of behaviors according to driving conditions, i.e., start-up, stopping, car-following, car-chasing and free driving. The classification criteria of driving behavior are shown in Table 4. According to the coupling relationship between driving behavior graph, the five types of behaviors are scored respectively.

According to the frequencies and rankings of the energy consumption pattern and the coding of patterns themselves, the ecology of the driving behavior was scored, as delivered in Eq. (4) as follows.

$$Score_x = 100 - 10 \times \frac{\sum_{i=1}^n \left(\frac{1}{3} \times \sum_{j=1}^3 Code_{ij} \times \frac{1}{Order_i} \right)}{\sum_{i=1}^n \frac{1}{Order_i}} \quad (4)$$

where $Score$ is the eco-driving score of a certain type of behavior; x is different driving behavior; $Code_{ij}$ is the code value of the j-second energy consumption pattern corresponding to the i-th driving pattern (For example, the code value of the third-second energy consumption pattern "345" is "5"), $Order_i$ is the frequency order of the i-th driving pattern, and

Table 3
Coding rules for energy consumption.

Energy consumption coding	Range of standardized average 100 km energy losses (EL)
0	-1.4~-0.7
1	-0.7~-0.5
2	-0.5~-0.3
3	-0.3~-0.1
4	-0.1~0.2
5	0.2~0.7
6	0.7~4.4

Table 4
Classification of driving behavior.

Type of driving	Driving behavior definition
Start-up	1) The local traffic state was CCCF or CSFF; 2) The first code of the driving behavior graph was ≥ 4 and the normalized speed ≤ -0.5 ; 3) The second and third codes of the driving behavior graph were ≥ 3 .
Stopping	1) The local traffic state was CCCF or CSFF; 2) The first code of the driving behavior graph was ≤ 2 ; 3) The second code of the driving behavior graph were ≤ 3 ; 4) The third code of the driving behavior graph were ≤ 3 and the normalized speed ≤ -0.5 .
Car-following	1) The local traffic state was CCCF or CSFF; 2) It didn't belong to the start-up and stopping.
Car-chasing	1) The local traffic state was CSFF; 2) It didn't belong to the start-up and stopping.
Free driving	1) The local traffic state was UFFF.

n is the number of typical driving patterns under this type of behavior in the acceleration graph of the driver. A lower score indicates more ecological driving behaviors.

5. Results and discussion

5.1. Driving behavior graph spectrum construction

5.1.1. Driving behavior graph spectrum portrait

Based on the graph spectrum construction method proposed above, the behavioral profiles of the 15 drivers were graphically demonstrated. Wherein, six drivers, including three mild driving style drivers which showing overall small code in graph (No.1, No.2, No.3) and three aggressive driving style drivers which showing overall large code in graph (No.13, No.14, No.15), were selected as examples for the driving behavior graph spectrum portrait (Fig. 7). As shown in Fig. 7, the local traffic states were labelled at the top of the graph, and the numbers left aside the graph presented the order of frequency of driving pattern in the corresponding local traffic status, and the color of each square stood for a particular code ("0" to "6" according to normalized acceleration) indicating a particular driving status. Take driver No. 1 as an example (Fig. 7(a)), under the congested close car-following (CCCF) traffic status, the top two most commonly seen driving pattern were "111" and "444", which means that drivers would more likely to continuously decelerate and accelerate a little in a short period in close following. This could be well in accordance with the actual "stop-and-go" traffic status of vehicles in congested situations. On the contrary, under the unconstrained fast free-flow (UFFF) conditions, there were much more green color appeared in Fig. 7(a), suggesting that the No. 1 driver had much more chance to keep a stabilized speed, i.e., the acceleration was zero (coded as "3" in green in the driving behavior graph in Fig. 7) while the traffic was in a free-flow states. Take the comparison among driver No. 2 and driver No. 3 as an example (Fig. 7(b) and (c)), the first six driving pattern in the CCCF traffic state were exactly the same and the all driving pattern in the UFFF traffic state were highly consistent, which indicated the driving behavior characteristics of the two drivers in extreme traffic state were highly similar. In the comparison among the three mild driving style drivers (No.1, No.2, No.3) and three aggressive driving style drivers (No.13, No.14, No.15), the most three frequent driving patterns under all traffic state were cruising (code "333"), slight deceleration (code "222") and slight acceleration (code "444") in mild driving style drivers, while the most frequent driving patterns under all traffic state were slight acceleration (code "444"), heavy acceleration (code "555") and rapid deceleration (code "000") in aggressive driving style drivers. This could be well in two different driving style drivers' behavior in real traffic state.

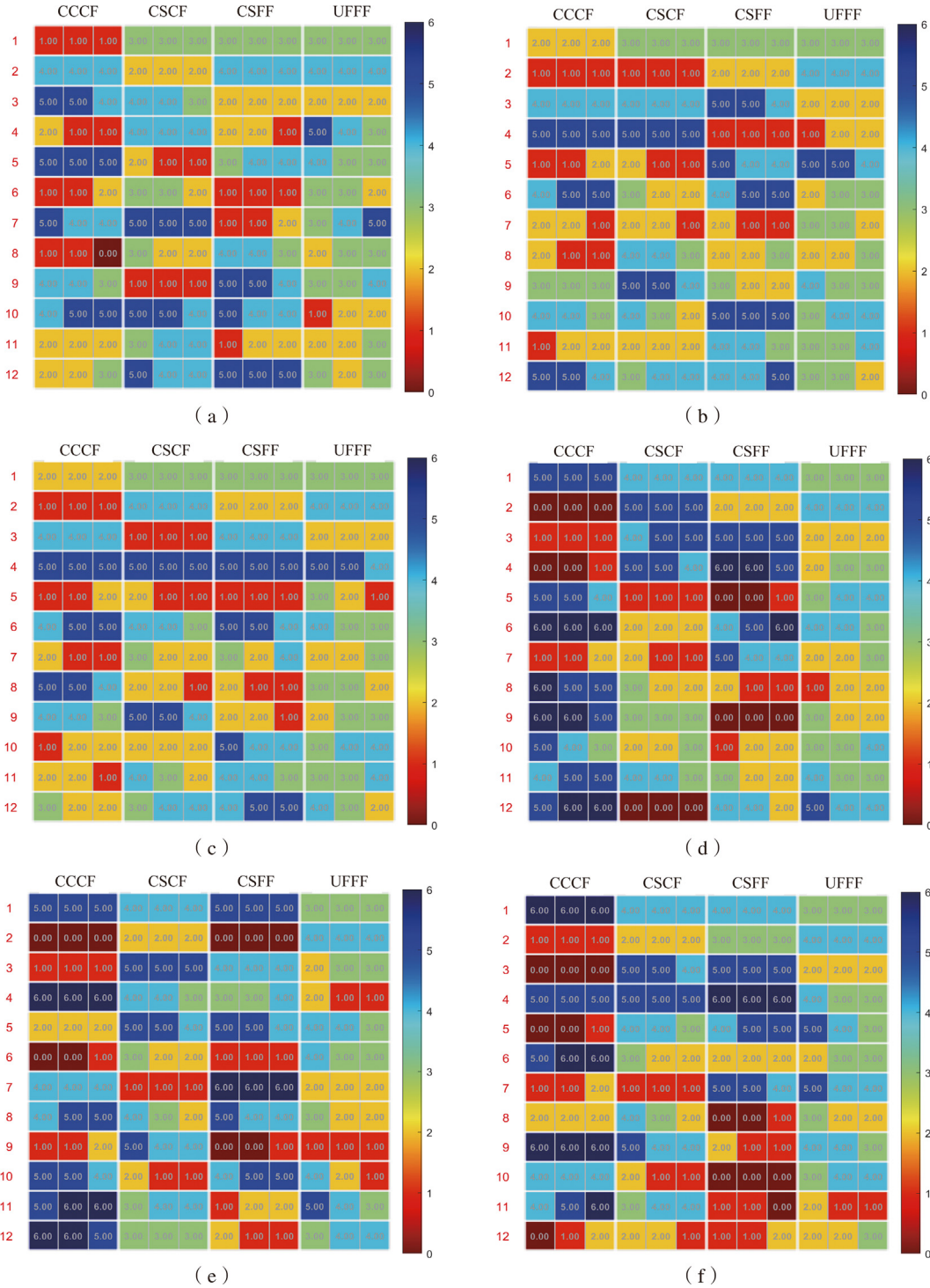


Fig. 7. Driving behavioral graph spectrums of selected drivers under varying traffic status. (a) Acceleration graph of NO.1 driver. (b) Acceleration graph of NO.2 driver. (c) Acceleration graph of NO.3 driver. (d) Acceleration graph of NO.13 driver. (e) Acceleration graph of NO.14 driver. (f) Acceleration graph of NO.15 driver.

5.1.2. Robustness analysis of diving behavior graph spectrums

While the driving behavioral graph spectrums could be constructed as above, but the graph spectrum of individual drivers may vary when the original driving behavioral data the graph derived from changes. So, the robustness of the driving behavioral graph spectrums was tested on varying available datasets.

This was achieved by taking account for the possible effects of the cumulative driving mileage on the performance of the driving behaviors. Since the cumulative driving mileage of the 15 drivers on the Second

Ring Road (non-tunnel section) for the whole experimental cycle (from 28 January 2022 to 7 December 2022) ranges from 613 to 1,086 km, the driving behavioral data of the 15 drivers were accumulated in 50, 75, 100, 110, ..., and 600 km to formulate various amount of driving behavioral datasets for graph spectrum construction separately (see Fig. 8). It's worth noting that the 600 km accumulated mileage was used to construct the standard driving behavior graph. It can be discovered from Fig. 8 that the driving behavior graphs of the individual drivers shared some similarities in terms of the accumulative effects of the

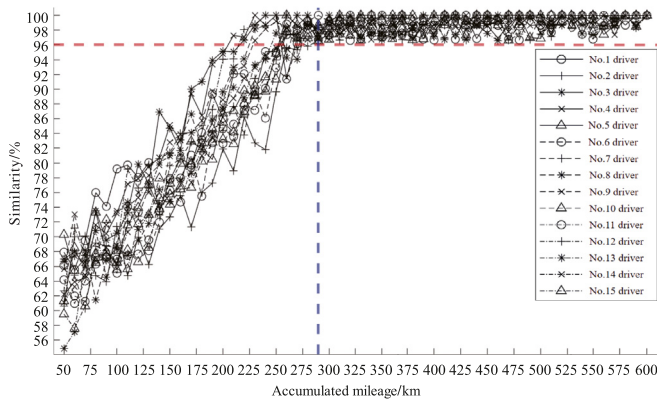


Fig. 8. Similarities of driving behavior graphs constructed over varying amount of driving behavioral data.

amount of the driving behavioral datasets. That was, the similarities between the constructed driving behavioral graphs increased from around 60% when the driving mileage was 50 km to around 96% when the mileage increased to 290 km, and then the similarities reached its limit without increase. It can be concluded that the driving behavior graph could be stabilized when the behavioral data were accumulated in at least 290 km. Accordingly, the behavioral data accumulated in 290 km were used to construct the driving behavior graph spectrum and energy consumption graph spectrum in following.

5.2. Energy consumption graph spectrum portrait and analysis of electric vehicles

According to the method introduced in Section 4.3, the energy consumption graph spectrums of individual drivers were portrayed. For the sake of clearly and simplification, the 3 drivers (No. 1, No. 2, and No. 3) with the lowest 100 km energy consumption and the 3 drivers (No.13, No.14, and No.15) with the highest 100 km energy consumption were particularly selected from the 15 drivers to demonstrate and compare the energy consumption graph spectrums. Fig. 9 shows the energy consumption graph spectrums of the six representative drivers under varying traffic status. The local traffic states were labelled at the top of the graph, and the numbers left aside the graph presented the order of frequency of energy consumption pattern in the corresponding local traffic status, and the color of each square stood for a particular code ("0" to "6" according to normalized energy losses) indicating a particular driving status. In Fig. 9, the most frequent codes under CCCF traffic state were code "5" and code "6", which indicated the energy consumption of vehicle "stop-and-go" behavior in CCCF traffic state was large. In comparison among the energy consumption of two categories drivers, code "222" appeared most frequently in the first pattern of low energy consumption drivers, while code "222" didn't appear in the first pattern of high energy consumption drivers. In comparison among the three high energy consumption drivers (No. 13, No. 14, and No. 15), driver No. 14 was relatively more ecology than the others based on the frequency of pattern '6' under all traffic state. And in this driver's graph, extreme energy patterns with larger or smaller codes were more distributed, which indicated the driver's driving behavior was less irregular in real complex traffic flow conditions.

The coupling analysis of the driving behavior graph spectrums (acceleration) and energy consumption graph spectrums of the six drivers was conducted respectively. Generally, based on Fig. 7 and 9, it can be discovered that the darker the squares in the driving behavior graph spectrums (Fig. 7) were, the darker the squares in the corresponding energy consumption graph spectrums (Fig. 9). The dark blue color of the driving behavior graph means rapid acceleration and the dark red color means rapid deceleration; the darker color of the energy consumption graph means higher energy consumption, which indicated the

correlation in the driving behavior graph and the energy consumption graph.

Take driver No. 13 as an example (Fig. 7(d) and 9(d)), all the rapid acceleration pattern (code "5" or code "6" only) or rapid deceleration pattern (code "0" or code "1" only) in driving behavior graph, showing the high energy consumption pattern (code "5" or code "6" only) in energy consumption graph. Take low energy consumption drivers (No. 1, No.2, No.3) as an example, the first driving pattern was same for both driving behavior graph and energy consumption graph under CSCF, CSFF, UFFF traffic state, which indicated the driving habit characteristics of the low energy consumption drivers in extreme traffic state were relatively similar.

In the comparison among the four different traffic states in two types of graph, the rapid acceleration pattern (code "5" or code "6" only), rapid deceleration pattern (code "0" or code "1" only) and high energy consumption pattern (code "5" or code "6" only) were more likely to occur in CCCF traffic state, this phenomenon was reverse in the UUFF traffic state. It indicated the drivers' driving habits in the two different traffic states were relatively inconsistent, this could be well in accordance with the actual traffic state in congested situations and free conditions. In the comparison among low energy consumption drivers (No. 1, No.2, No.3) and high energy consumption drivers (No. 13, No.14, No.15) in energy consumption graph, the three high energy consumption drivers' driving behavior graph occurred the rapid acceleration pattern (code "666"), rapid deceleration pattern (code "000"), while the pattern didn't occur in the three low energy consumption drivers' driving behavior graph, which indicated the severely rapid acceleration or deceleration would increase the energy consumption. In addition, it was found that code "3" occurred more frequently in the driving behavior graph of low energy consumption drivers, indicating that the "cruising" pattern was more ecological.

Take a particular driving pattern (code "111") in driving behavior graph as an example. This driving pattern indicated moderate deceleration, which had low energy consumption (all numbers of the code were code "0" or "1") in CSCF, CSFF and UFFF traffic state. However, this driving pattern had high energy consumption (all numbers of the code were not "0" or "1") in CCCF traffic state, while the energy consumption code increased sequentially, and even high energy consumption occurred in the third second. This phenomenon was related to the energy feedback characteristics of electric vehicles, energy recovery occurred when the vehicle was decelerating, the degree of energy recovery decreases sharply under the rapid deceleration, and the percentage of energy dissipation due to braking, etc., increases dramatically, which lead to the energy consumption change of the pattern (code "111") in the CCCF traffic state.

5.3. Quantitative score of ecology driving behavior

Using the quantitative analysis algorithm described in Section 4.4, the driving behavior ecological grades of the fifteen drivers were calculated, a lower score indicates more ecological driving behaviors. Based on the previous driving behavior graph and the four traffic states, five types of behaviors were taken into consideration: start-up, stopping, car-following, car-crashing and free-driving, the ecological scores were shown in Table 5. The scores can effectively distinguish between ecological of fifteen drivers, and the tendency of the scores was generally in accordance with the division of the three high energy consumption drivers and three low energy consumption drivers in previous work.

According to the calculated ecological scores of different driving behaviors of fifteen drivers, the boxplot of different driving behaviors of fifteen drivers as shown in Fig. 10.

As can be seen from Table 5 and Fig. 10, take the start-up, stopping and free driving as an example, the scores of start-up were generally low, such scores could further support the conclusion of previous analysis that rapid acceleration, rapid deceleration and "stop-and-go" behavior in congested conditions may increase energy consumption, this result ties well with previous studies [20,22,23]. Meanwhile, the scores of free driving were high, which was in contrast to the scores of start-up. The

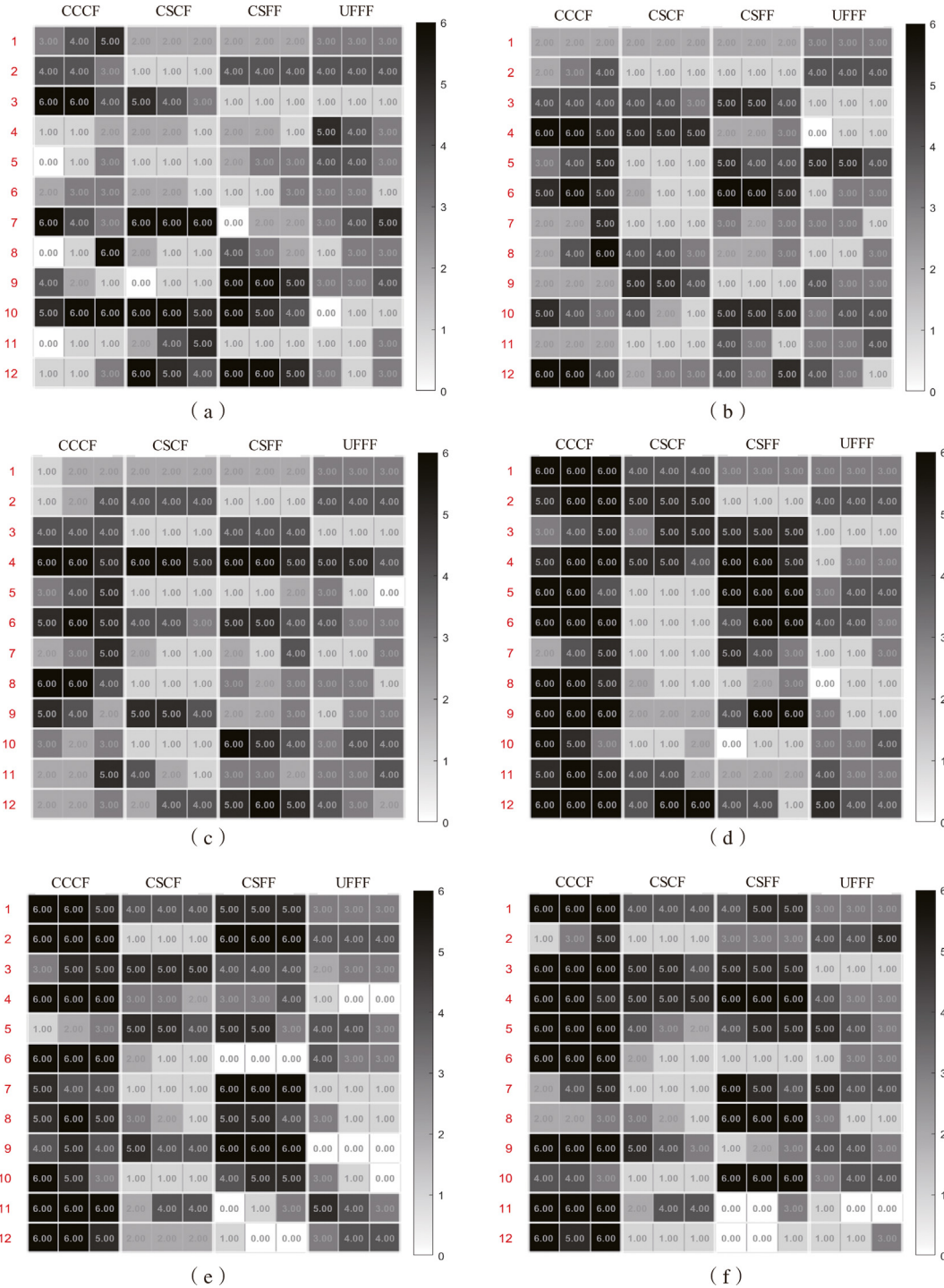


Fig. 9. Energy consumption graphs of varying drivers. (a) Energy consumption graph for NO.1 driver. (b) Energy consumption graph for NO.2 driver. (c) Energy consumption graph for NO.3 driver. (d) Energy consumption graph for NO.13 driver. (e) Energy consumption graph for NO.14 driver. (f) Energy consumption graph for NO.15 driver.

results can be well in line with the actual traffic states under congestion and free conditions. In addition, the score of stopping behavior was in the middle, which was also reasonable, because the energy consumption of stopping behavior was affected by the speed and acceleration in the period before stopping.

In the comparison among the car-following, car-chasing. The maximum score of car-following and car-chasing was similar, which was the highest score. However, the minimum score of car-chasing was lower, and widely dispersed. Combining the definition of five behavior and

comparing the four traffic states, a careful examination of the cause of this result revealed that the distribution degree of different driving patterns in CSFF state was relatively scattered. Although the similar distribution was occurred in the CSCF traffic state, the UFFF state was taken into account in the car-following, which lead to the difference in the scores of the two behaviors. Meanwhile, the low energy consumption pattern (code "0" or code "1" only) occurred in the three traffic states in the energy consumption graph, which can be well in line with results of the maximum scores of these three behaviors tend to be consistent.

Table 5

Drivers' score of the eco-driving performance.

Number of drivers	Average energy consumption /(kwh·(100 km) ⁻¹)	Score of eco-driving				
		Start-up	Stopping	Car-following	Car-chasing	Free driving
1	13.38	48.71	77.88	72.12	74.79	71.21
2	14.25	46.67	72.69	73.69	75.17	72.20
3	14.51	45.00	68.07	74.29	76.09	71.22
4	14.91	40.00	56.74	68.19	82.56	71.19
5	15.20	59.44	63.08	57.89	65.43	67.76
6	15.71	41.96	51.71	68.20	63.10	74.32
7	15.94	42.86	60.00	71.22	57.93	65.60
8	16.19	44.56	49.18	63.42	55.64	81.20
9	16.31	40.00	46.98	73.71	60.57	70.11
10	16.67	43.19	46.93	72.60	52.85	72.19
11	16.97	50.59	48.61	57.97	58.27	70.45
12	17.26	43.33	48.60	63.17	56.44	72.54
13	17.46	41.67	47.03	56.05	66.98	71.75
14	17.77	41.37	41.33	64.51	54.01	72.70
15	18.10	40.00	52.54	68.72	40.00	70.05

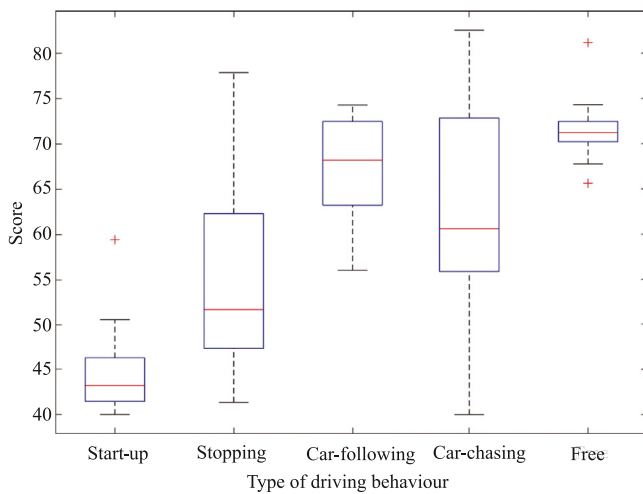


Fig. 10. The box-plot of eco-driving score.

6. Conclusion

In this paper, a graphical representation method was proposed to construct a graphical model of individual driving behavior. This method visualizes the driver's driving behavior on specific road types by extracting and sequencing the typical driving patterns of drivers in various local traffic states. As a graph method, driving behavior graph can not only directly reveal the driving habits of drivers by describing the individual behavior characteristics of different drivers, but also obtain the energy consumption characteristics of drivers' driving behaviors through coupling analysis with energy consumption. The results showed: 1) The severely rapid acceleration or deceleration would increase the energy consumption, and it is recommended that drivers try to accelerate as slowly as possible in all traffic conditions, avoiding rapid acceleration patterns to achieve a more ecological driving behavior; 2) The driving behavior was more complex and un-ecological in CCCF traffic state, this could be well in accordance with the actual traffic state; 3) The energy consumption change of a particular driving pattern (code "111") under CCCF traffic state was relatively affected to the energy feedback characteristics of electric vehicles; 4) five different driving behaviors produced different driving energy consumption, and thus caused a large difference in ecology.

In addition, a quantitative analysis method of driving behavior ecology based on the coupling of driving behavior graph, energy consumption graph was also provided to quantify the difference between drivers' driving behavior ecology, and an ecological score graph was

formed to clearly express the ecological behavior of driving and put forward personalized ecological driving suggestions for drivers. By studying various ecological components and the relationships among them, the ecological nature of drivers' behaviors and the relationships among them are explored, and the influence mechanism of ecological driving behaviors of electric vehicles is analyzed, which lays a theoretical foundation for the realization of ecological driving regulation. However, this research only analyzed the ecological driving behavior of electric vehicles. In the future, as to propose more targeted ecological driving regulation strategies, it is necessary to compare the differences in energy saving performance of typical driving behaviors under fuel vehicles and electric vehicles.

CRedit authorship contribution statement

Hui Zhang: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Yiyue Luo:** Writing – original draft, Validation, Formal analysis, Data curation. **Naikan Ding:** Writing – review & editing, Writing – original draft, Validation. **Toshiyuki Yamamoto:** Writing – review & editing. **Chenming Fan:** Visualization, Software, Methodology, Investigation. **Chunhui Yang:** Validation, Data curation. **Wei Xu:** Validation, Data curation. **Chaozhong Wu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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